Introduction to Real Analysis – Homework 8

Abstract integration and monotone convergence

Let $(X, \mathcal{A}, \mu)$ be a measure space. Use $0 \cdot \infty = 0$; m denotes Lebesgue measure.

1. **Truncation.** Let $f : X \to \mathbb{R}$ be integrable and set

$$f_n = \max\{-n, \min\{f, n\}\}.$$

Show that each f_n is integrable and that

$$\int_X |f - f_n|\, d\mu \longrightarrow 0, \qquad \int_X f_n\, d\mu \longrightarrow \int_X f\, d\mu.$$

Hint. Apply monotone convergence to $\min\{|f|, n\}$.

2. **Nonnegative series.**

(a) If $u_k : X \to [0, \infty]$ are measurable, prove that $\sum_{k \geq 1} u_k$ is measurable and

$$\int_X \sum_{k=1}^{\infty} u_k\, d\mu = \sum_{k=1}^{\infty} \int_X u_k\, d\mu.$$

(b) Compute $\int_{\mathbb{R}} h\, dm$, where

$$h(x) = \sum_{k=1}^{\infty} \mathbf{1}_{(0,\, 2^{-k})}(x).$$

3. **Decreasing sequences.**

(a) Let $f_n : X \to [0, \infty)$ be measurable, with $f_n \downarrow f$ pointwise and $\int_X f_1\, d\mu < \infty$. Prove that

$$\int_X f_n\, d\mu \longrightarrow \int_X f\, d\mu.$$

Hint. Consider $f_1 - f_n$.

(b) Give a counterexample on $\mathbb{R}$ with Lebesgue measure if the assumption $\int_X f_1\, d\mu < \infty$ is removed.

4. **A measure defined by an integral.** Let $w : X \to [0, \infty]$ be measurable and define $\nu(E) = \int_E w\, d\mu$ for $E \in \mathcal{A}$.

(a) Prove that ν is a measure on $(X, \mathcal{A})$.

(b) For every measurable $f : X \to [0, \infty]$, prove that fw is measurable and

$$\int_X f\, d\nu = \int_X fw\, d\mu.$$

Hint. Begin with bounded nonnegative simple functions, then approximate from below. Problem 2(a) may be used in part (a).